

Aster Pump X17

User Guide and Troubleshooting Manual



1. Product Overview

The Aster Pump X17 is a fictional smart cooling pump used for this local aftercare and RAG proof of concept. The unit supports three operating modes: Bluefin, Orchard, and Nightglass.

- Bluefin mode is used for normal daytime cooling.
- Orchard mode is used only when humidity is above 72 percent.
- Nightglass mode is used for quiet overnight operation.

2. Operating Modes

Mode	Dial Position	When to Use	Important Restriction
Bluefin	4	Normal daytime cooling	Default mode for stable operation.
Orchard	6	Humidity above 72 percent	Do not use below 72 percent humidity.
Nightglass	2	Silent overnight operation	Do not use during pressure echo recovery.

3. Error Codes

E-41: Silver Impeller Sensor Misalignment

Power down the unit, reseal the silver impeller sensor, and run calibration cycle C2 before returning the unit to service.

E-77: Coolant Loop Pressure Echo

Error code E-77 means the coolant loop has detected a pressure echo. Inspect the return valve, drain 200 ml of coolant, and restart the pressure monitor.

If E-77 appears twice within one hour, do not restart the unit. Escalate to a Level 2 technician and leave the pressure monitor offline.

E-93: Filter Cassette Expired

Replace the filter cassette and confirm the cassette date stamp in the service panel before restarting normal operation.

4. Maintenance Schedule

- Replace the filter cassette every 19 days, not monthly.
- Write the replacement date on the green service tag.
- Inspect the coolant loop every 11 days.
- Run calibration cycle C2 after sensor replacement, moving the unit more than 5 meters, or firmware updates.
- Only Level 2 technicians may approve operation after repeated E-77 pressure echo events.